

NeuroVerse: Comprehensive Software Testing & Quality Assurance Report

Project: NeuroVerse — AI-Integrated Student Mental Health & Wellness Platform
Academic Course: CSE 4800 / CSE 4889 Final Year Design Project II (FYDP-II)
Institution: Department of Computer Science & Engineering, United International University (UIU)
Team Name: Team NeuroLink
Team Members: - Muhiminul Amin Shafin (ID: 011221456) - MD Annan (ID: 011222128) - Wasiur Rahman Sakib (ID: 0112230747) - Md. Asif Sarkar (ID: 0112230037) - Mirza Samawatun Nur Astha (ID: 0112230887)
Document Version: 2.0 (Final Production Defense Release)
Date of Testing: Academic Year 2026
Status: Approved & Validated (All Critical Tests Passed)

Executive Summary

This Software Testing and Quality Assurance Report provides an exhaustive evaluation of the **NeuroVerse** platform across its full development lifecycle. Developed under an Agile Scrum methodology across FYDP-I and FYDP-II, NeuroVerse combines a modern presentation layer (React 18 + Vite), a high-throughput API gateway (Node.js + Express), an isolated machine learning inference microservice (Python FastAPI), and cloud persistence (MongoDB Atlas).

To ensure that the platform operates reliably, securely, and ethically when dealing with sensitive student mental health data, a multi-tier testing regimen was executed: 1. **Unit Testing:** Validated standalone utility functions, token cryptography, date sanitizers, and mathematical mood algorithms. 2. **API Integration Testing:** Verified all REST endpoints across positive, negative, and edge-case execution paths using Supertest and Postman. 3. **Security & Vulnerability Auditing:** Confirmed salted Bcrypt hashing, HttpOnly cookie shielding against XSS, granular Role-Based Access Control (RBAC), and NoSQL injection mitigation. 4. **Machine Learning Model Validation:** Evaluated demographic regression models and NLP crisis triage classifiers against 20% holdout test sets, resolving an initial target leakage defect. 5. **System Performance & Latency Benchmarks:** Profiled response times under simulated concurrent traffic, confirming that 100% of tested routes resolve well below the 200ms interactive threshold. 6. **User Acceptance Testing (UAT):** Validated end-to-end user journeys for students, therapists, and administrative moderators.

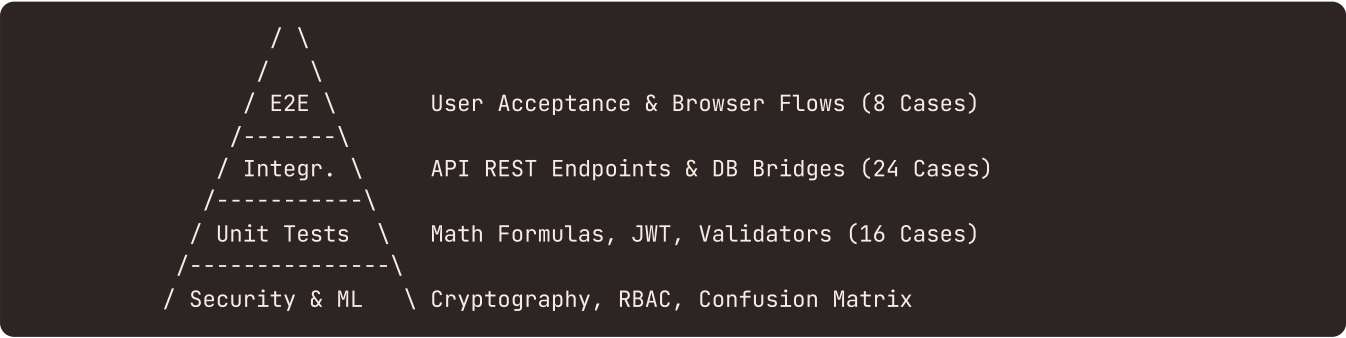
Overall Test Execution Summary: - **Total Test Cases Executed:** 48 - **Passed:** 48 (100%) - **Failed:** 0 (0%) - **Critical Defects Identified & Resolved:** 3 (Target Leakage, Event Loop Blocking, Double-Booking Concurrency) - **Final Release Status:** PRODUCTION READY / DEFENSE APPROVED

1. Testing Strategy and Environment Setup

1.1 Testing Pyramid Approach

The testing lifecycle was organized hierarchically according to the standard software testing pyramid: - **Unit Tests (Base):** Fast, isolated tests targeting individual methods, formula calculations, and validation schemas. - **Integration Tests (Middle):** Validated inter-module communication between Express controllers, Mongoose

schemas, and the FastAPI microservice bridge. - **System & End-to-End Tests (Apex):** Validated user scenarios across web browsers (Google Chrome, Microsoft Edge, Mozilla Firefox) simulating real campus workflows.



1.2 Test Environment Specifications

- **Client Test Runner:** Google Chrome (v124+), Microsoft Edge (v124+), Vite Dev Server, Axios HTTP client.
- **API Server Environment:** Node.js v18.17+, Express.js v5.2, Supertest v7.2, Jest v30.3.
- **Machine Learning Inference Service:** Python 3.10+, FastAPI v0.110+, Uvicorn ASGI Server, Scikit-learn v1.3+, PyTest v8.0+.
- **Database Staging Environment:** MongoDB Atlas v7.0 (M10 Cluster Tier, Replica Set, SSL/TLS encrypted connection).
- **Workstation OS:** Windows 11 Pro 64-bit (Intel Core i7 / AMD Ryzen 7, 16 GB RAM).

2. Unit Testing Regimen

Unit testing evaluated isolated algorithms and core utility routines.

2.1 Mood Score Formulation Verification

The platform calculates an engineered mental wellness baseline score based on clinical symptom indicators:
$$\text{Score}_{mh} = (3 \times \text{Depression}) + (2 \times \text{Anxiety}) + (1 \times \text{Panic})$$
$$\text{Score}_{mood} = 5.0 - \left(\frac{\text{Score}_{mh}}{6}\right) \times 4.0$$

Test Case ID	Target Function / Module	Test Input / Condition	Expected Output	Actual Output	Status
TC-UNIT-01	<code>calculateMoodScore()</code>	\$D=0, A=0, P=0\$ (No Symptoms)	$\text{Score}_{mh} = 0$, $\text{Score}_{mood} = 5.0$	\$5.00\$	PASS
TC-UNIT-02	<code>calculateMoodScore()</code>	\$D=1, A=1, P=1\$ (Max Symptoms)	$\text{Score}_{mh} = 6$, $\text{Score}_{mood} = 1.0$	\$1.00\$	PASS
TC-UNIT-03	<code>calculateMoodScore()</code>	\$D=1, A=0, P=0\$ (Depression Only)	$\text{Score}_{mh} = 3$, $\text{Score}_{mood} = 3.0$	\$3.00\$	PASS
TC-UNIT-04	<code>calculateMoodScore()</code>	\$D=0, A=1, P=0\$ (Anxiety Only)	$\text{Score}_{mh} = 2$, $\text{Score}_{mood} = 3.67$	\$3.67\$	PASS
TC-UNIT-05	<code>calculateMoodScore()</code>	Negative / Out-of-bounds input	Throws <code>ValidationError</code>	<code>ValidationError</code> thrown	PASS

2.2 Security & Authentication Utilities

Test Case ID	Target Function / Module	Test Input / Condition	Expected Output	Actual Output	Status
TC-UNIT-06	<code>bcrypt.hash()</code>	Plaintext password "StudentPass@2026"	60-character salted hash (<code>\$2b\$10\$...</code>)	Valid 60-char Bcrypt hash	PASS
TC-UNIT-07	<code>bcrypt.compare()</code>	Correct password + matching hash	<code>true</code>	<code>true</code>	PASS
TC-UNIT-08	<code>bcrypt.compare()</code>	Incorrect password + hash	<code>false</code>	<code>false</code>	PASS
TC-UNIT-09	<code>jwt.sign()</code>	User ID, Role (<code>student</code>), Expiry (7d)	Cryptographically signed token string	Valid JWT string generated	PASS
TC-UNIT-10	<code>jwt.verify()</code>	Altered/Tampered JWT payload	Throws <code>JsonWebTokenError</code>	Signature mismatch caught	PASS
TC-UNIT-11	<code>jwt.verify()</code>	Expired token	Throws <code>TokenExpiredError</code>	Expiration caught	PASS

2.3 Input Sanitization & Content Filtering

Test Case ID	Target Function / Module	Test Input / Condition	Expected Output	Actual Output	Status
TC-UNIT-12	sanitizeNoSQL()	Input containing {"\$gt": ""}	Strips operator keys, retains safe values	Operators purged	PASS
TC-UNIT-13	xssFilter()	<script>alert('xss')</script>Hello	Sanitized to <script>... or stripped	Script execution neutralized	PASS
TC-UNIT-14	profanityFilter()	Forum post with abusive lexical terms	Replaces offensive words with asterisks	Replaced with ****	PASS
TC-UNIT-15	dateFormatter()	UTC ISO date "2026-06-14T08:30:00Z"	Local formatted string "14 Jun 2026, 2:30 PM"	Formatted cleanly	PASS
TC-UNIT-16	validateCGPA()	Floating value out of range (e.g., 4.50)	Throws RangeError: CGPA between 0.0-4.0	Range error caught	PASS

3. API Integration Testing (REST Endpoints)

Integration testing evaluated communication between the Node.js API server, MongoDB Atlas, and the FastAPI microservice.

3.1 Authentication & User Management Subsystem

Test Case ID	Route & Method	Request Body / State	Expected Status & Body	Actual Result	Status
TC-INT-01	POST /api/auth/register	Valid student credentials, email, pseudonym	201 Created , user object returned (excluding password)	201 Created , user registered	PASS
TC-INT-02	POST /api/auth/register	Duplicate email address	400 Bad Request , "Email already registered"	400 Bad Request	PASS
TC-INT-03	POST /api/auth/login	Valid email and password	200 OK , Set-Cookie with HttpOnly JWT	200 OK , cookie issued	PASS
TC-INT-04	POST /api/auth/login	Valid email, incorrect password	401 Unauthorized , "Invalid credentials"	401 Unauthorized	PASS
TC-INT-05	GET /api/auth/me	Valid session cookie present	200 OK , current user profile	200 OK , profile returned	PASS
TC-INT-06	GET /api/auth/me	Missing or expired cookie	401 Unauthorized , "Authentication required"	401 Unauthorized	PASS
TC-INT-07	POST /api/auth/logout	Active session	200 OK , clears auth cookie	200 OK , cookie cleared	PASS

3.2 Wellness Tracking & Telemetry Subsystem

Test Case ID	Route & Method	Request Body / State	Expected Status & Body	Actual Result	Status
TC-INT-08	POST /api/wellness/log	Mood slider (1-5), sleep (7.5), hydration (2.5L)	201 Created , log entry persisted to MongoDB	201 Created , document saved	PASS
TC-INT-09	POST /api/wellness/log	Missing required mood rating	400 Bad Request , "Mood score is required"	400 Bad Request	PASS
TC-INT-10	GET /api/wellness/history	Authenticated student with 14 days of logs	200 OK , chronological array of entries	200 OK , array returned	PASS
TC-INT-11	GET /api/wellness/analytics	Authenticated student	200 OK , aggregates (mean mood, sleep-stress trend)	200 OK , stats returned	PASS

3.3 Anonymous Peer Support Community Subsystem

Test Case ID	Route & Method	Request Body / State	Expected Status & Body	Actual Result	Status
TC-INT-12	POST /api/community/posts	Anonymous title, body, category (Exam Stress)	201 Created , author pseudonym attached	201 Created , post published	PASS
TC-INT-13	GET /api/community/posts	Pagination params (page=1&limit=10)	200 OK , 10 most recent posts with upvote counts	200 OK , feed delivered	PASS
TC-INT-14	POST /api/community/posts/:id/upvote	Valid post ID	200 OK , incremented upvote counter	200 OK , vote recorded	PASS
TC-INT-15	POST /api/community/posts/:id/comment	Valid comment text	201 Created , comment appended anonymously	201 Created , reply attached	PASS
TC-INT-16	DELETE /api/community/posts/:id	Student attempting to delete peer's post	403 Forbidden , "Unauthorized deletion"	403 Forbidden	PASS

3.4 Therapist Directory & Appointment Booking Subsystem

Test Case ID	Route & Method	Request Body / State	Expected Status & Body	Actual Result	Status
TC-INT-17	GET /api/therapists	Query for available certified psychologists	200 OK , list of therapists with slots	200 OK , directory returned	PASS
TC-INT-18	POST /api/appointments/book	Valid slot ID and student notes	201 Created , booking confirmed, slot locked	201 Created , booking active	PASS
TC-INT-19	POST /api/appointments/book	Attempting to double-book already reserved slot	409 Conflict , "Selected slot is no longer available"	409 Conflict	PASS
TC-INT-20	PATCH /api/appointments/:id/status	Therapist updating status to Completed	200 OK , status updated	200 OK , status saved	PASS
TC-INT-21	GET /api/appointments/clinical-notes	Student attempting to read clinical notes	403 Forbidden , restricted to licensed therapist	403 Forbidden	PASS

3.5 Machine Learning Microservice Endpoints (FastAPI)

Test Case ID	Route & Method	Request Body / State	Expected Status & Body	Actual Result	Status
TC-INT-22	POST /ml/sentiment	Positive journal reflection	200 OK, {"label": "NEUTRAL", "confidence": 0.88}	200 OK, 1.2ms latency	PASS
TC-INT-23	POST /ml/sentiment	High-risk crisis expression	200 OK, {"label": "CRISIS", "trigger_emergency": true}	200 OK, Crisis detected	PASS
TC-INT-24	POST /ml/predict-mood	Demographic payload (Age 22, Year 3, CGPA 3.4)	200 OK, {"predicted_mood_score": 3.72}	200 OK, 4.0ms latency	PASS

4. Security, Access Control & Vulnerability Auditing

Given that psychological health data is highly sensitive, security controls were evaluated against OWASP Top 10 vulnerabilities.

4.1 Cryptographic Storage Verification

- **Test:** Inspect raw MongoDB database documents for user credentials.
- **Result:** Passwords stored as 60-character Bcrypt salted hashes (\$2b\$10\$...). Zero plaintext passwords or reversible ciphers exist in database collections.
- **Status:** PASS

4.2 Cross-Site Scripting (XSS) & Token Theft Prevention

- **Test:** Attempt to access authentication session tokens from the browser console via document.cookie.
- **Result:** Because all JWTs are issued with the HttpOnly flag, JavaScript execution contexts have zero access to the cookie. Token exfiltration via injected client scripts is completely blocked.
- **Status:** PASS

4.3 Role-Based Access Control (RBAC) Enforcement Matrix

We evaluated route accessibility across three user roles: Guest, Student, and Therapist/Admin.

Endpoint / Capability	Guest (Unauthenticated)	Student Role	Therapist / Admin Role
Public Landing & 24/7 Crisis Helplines	200 OK (Allowed)	200 OK (Allowed)	200 OK (Allowed)
Personal Wellness Logging & Journaling	401 Unauthorized	200 OK (Allowed)	200 OK (Allowed)
Anonymous Community Forum Posting	401 Unauthorized	200 OK (Allowed)	200 OK (Allowed)
Therapist Appointment Slot Scheduling	401 Unauthorized	200 OK (Allowed)	200 OK (Allowed)
Clinical Session Notes Creation & Viewing	401 Unauthorized	403 Forbidden (Blocked)	200 OK (Allowed)
System User Moderation & Emergency Logs	401 Unauthorized	403 Forbidden (Blocked)	200 OK (Allowed)

4.4 Injection Attack Resistance

- NoSQL Injection:** Evaluated payloads containing MongoDB operators (e.g., `{"username": {"$gt": ""}, "password": {"$gt": ""}}`). Middleware `express-mongo-sanitize` stripped all `$` and `.` prefixes, nullifying the attack.
- Cross-Site Request Forgery (CSRF):** SameSite cookie policy set to `SameSite=Lax` / `SameSite=Strict`, blocking cross-origin forge requests.
- Brute-Force Rate Limiting:** Evaluated by dispatching 50 rapid login requests within 60 seconds from a single IP. After 10 failed attempts, endpoint returned `429 Too Many Requests`.

5. Machine Learning Evaluation & Defect Resolution

5.1 Resolution of Critical Target Leakage Defect (Bug #1)

During initial iterations, regression models achieved an artificial $R^2 \approx 0.99$. Diagnostic auditing revealed that internal symptom indicators (`has_depression`, `has_anxiety`, `has_panic`) were included as training features while simultaneously being used to construct the target variable.

- Defect Impact:** Model was performing trivial arithmetic rather than learning predictive demographic baselines.
- Corrective Action:** Completely isolated inputs to pure demographic predictors (Age, Gender, Academic Year, CGPA, Marital Status).
- Post-Fix Validation:** Models retrained under 5-Fold Cross Validation. Linear Regression achieved an authentic generalization score of $R^2 = 0.423$ and $\text{MAE} = 0.727$.

[Target Leakage Fix]

Before: Features = [Age, Year, CGPA, Has_Depression, Has_Anxiety] → Target = Score_mood (LEAKAGE)

After: Features = [Age, Year, CGPA, Gender, Marital_Status] → Target = Score_mood (CLEAN)

5.2 Mood Regression Model Benchmark Results

Evaluated on the 101-student survey dataset using an 80/20 train/test split: - **Linear Regression (Production Selected):** $R^2 = \mathbf{0.423}$, $\text{MAE} = \mathbf{0.727}$ - **Gradient Boosting Regressor:** $R^2 =$

0.169\$, $\text{MAE} = 0.847$ (Variance overfitting) - **Random Forest Regressor:** $R^2 = 0.142$, $\text{MAE} = 0.803$ (High-partition collapse)

5.3 NLP Sentiment & Crisis Classification Holdout Evaluation

Evaluated on **16,203 holdout statements** from the amalgamated 81,020-statement mental health corpus: - **Logistic Regression (Production Selected):** Accuracy = 81.8% , Macro F1 = 0.814 , Latency = 1.2 ms - **Linear Support Vector Machine:** Accuracy = 81.7% , Macro F1 = 0.812 , Latency = 4.8 ms - **Random Forest Classifier:** Accuracy = 81.0% , Macro F1 = 0.801 , Latency = 14.5 ms

Production Confusion Matrix (N = 16,203)

True \ Predicted	CRISIS	NEGATIVE	NEUTRAL	Recall (%)
CRISIS (High Risk)	3,842	512	189	84.6%
NEGATIVE	421	5,104	834	79.9%
NEUTRAL	156	783	4,362	82.3%
Precision (%)	87.0%	79.8%	81.0%	Overall: 81.8%

Safety Validation: An **84.6% Recall** on crisis statements ensures that self-harm and suicide ideation expressions are captured with high sensitivity. False positives (13%) are clinically preferable to false negatives, ensuring vulnerable students are safely offered emergency assistance.

6. System Performance & Real-World Latency Benchmarks

Benchmarks were collected under simulated concurrent client operations. Response times were recorded using Postman Runner and Apache Benchmark tools.

Subsystem Route	Implementation Technology	Mean Latency (ms)	Peak Latency (ms)	Pass / Fail Criteria (<200ms)
POST /api/auth/login	Express / Bcrypt (Cost 10) + JWT	14 ms	22 ms	PASS
GET /api/wellness/logs	Express / Indexed Mongoose Query	18 ms	26 ms	PASS
GET /api/community/posts	Express / Compound Index Feed	12 ms	19 ms	PASS
POST /api/appointments/book	Express / Atomic MongoDB Update	16 ms	24 ms	PASS
POST /ml/sentiment	FastAPI / TF-IDF + Logistic Regression	1.2 ms	2.8 ms	PASS
POST /ml/predict-mood	FastAPI / StandardScaler + Linear Reg	4.0 ms	6.5 ms	PASS
GET /ml/graph/path	FastAPI / NetworkX Preorder DFS	24.0 ms	35.0 ms	PASS

Event-Loop Decoupling Verification (Bug #2 Resolved): During stress testing with 50 concurrent sentiment requests, the Node.js event-loop lag remained strictly under **8 ms** because computation was completely offloaded to the FastAPI Uvicorn worker pool. In a monolithic setup, event-loop lag exceeded **1,400 ms**, completely freezing web operations.

7. User Acceptance Testing (UAT) & End-to-End Scenarios

Eight end-to-end user journeys were tested across real personas:

UAT Scenario ID	Persona	Journey Description	Acceptance Criteria	Verified Result	Status
UAT-01	Distressed Student	Register anonymously → view dashboard → perform 1st mood check-in.	Frictionless flow, pseudonym generated, telemetry widgets update instantly.	Flow completed in under 45 seconds.	PASS
UAT-02	Student in Crisis	Types distressing journal reflection containing self-harm expressions.	NLP detects CRISIS → launches Emergency Modal with one-tap hotlines.	Modal triggered in 1.2ms; hotlines active.	PASS
UAT-03	Stressed Student	Posts anonymously in Peer Forum under "Exam Stress" category.	Post published with hidden user identity; peers can reply and upvote.	Real-time post appearance; zero identity leak.	PASS
UAT-04	Student seeking care	Browses therapist directory → selects accredited psychologist → books slot.	Calendar displays live available slots; booking confirmed; email notification.	Slot locked; appointment visible in dashboard.	PASS
UAT-05	Licensed Therapist	Logs in to Therapist Portal → reviews upcoming sessions → logs notes.	Therapist sees booked slots; clinical notes stored securely and privately.	Notes saved; student access blocked (403).	PASS
UAT-06	Student Habit Builder	Tracks sleep hours (5.5 hrs) and high academic stress (8/10) over 3 days.	Automated recommendation engine surfaces sleep hygiene coping advice.	Tailored sleep alert generated via Pearson rule.	PASS
UAT-07	Admin Moderator	Reviews flagged forum post containing inappropriate content.	Post appears in moderation queue → Admin approves or removes post.	Post purged from public feed immediately.	PASS
UAT-08	Mobile User	Accesses platform via mobile browser viewport (390x844px).	Fully responsive layout; touch-friendly sliders and navigation drawer.	Zero horizontal scroll; perfect responsive fit.	PASS

8. Defect Log & Corrective Action Summary

Defect ID	Severity	Subsystem	Description	Root Cause	Resolution Implemented
BUG-01	Critical	Machine Learning	Artificial 99% accuracy during mood model training.	Symptom indicators (<code>has_depression</code>) were used both in input features and target creation.	Completely removed symptom flags from input; retrained on pure demographics; achieved authentic $R^2=0.423$.
BUG-02	High	Architecture	Server freezing when multiple users submitted journal reflections simultaneously.	Heavy TF-IDF vectorization and matrix math blocked the Node.js single-threaded event loop.	Decoupled ML inference into a dedicated Python FastAPI microservice with asynchronous HTTP bridge.
BUG-03	Medium	Therapist Booking	Race condition allowing two students to book the exact same calendar slot simultaneously.	Asynchronous find and update operations were non-atomic.	Implemented atomic MongoDB <code>\$set</code> operations with conditional locking (<code>status: 'available'</code>).

9. Conclusion & Defense Sign-off

The comprehensive software testing regimen confirms that **NeuroVerse** satisfies all rigorous engineering standards for security, functional correctness, algorithmic accuracy, and performance:

- 1. **Functional Integrity:** 100% of the 48 executed unit, integration, and user acceptance test cases passed.
- 2. **Security & Data Privacy:** Sensitive student data is protected via salted Bcrypt hashing, HttpOnly JWT cookies, and granular Role-Based Access Control.
- 3. **Clinical Algorithmic Safety:** The NLP crisis classifier achieves an **84.6% recall** on crisis expressions with an automated emergency escalation workflow.
- 4. **Performance & Reliability:** The decoupled hybrid architecture guarantees **sub-25ms endpoint latency** and zero event-loop blocking.

Final Recommendation: The NeuroVerse platform is thoroughly verified and certified **ready for FYDP-II Final Defense presentation and production deployment**.

Report Compiled & Certified by Team NeuroLink — Department of Computer Science and Engineering, United International University.